WORKED SOLUTION

STEP III 2022, Question 9

Mechanics •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Every collision here is the same collision: a moving particle hits a stationary one k times as heavy. So the factors α and β from the start of part (i) can be reused in part (ii), where B (mass km) hits C (mass k^2m). After that, each "meet at the same point" condition is just two expressions for the same time set equal to each other. In part (ii) we get two equations for k and e, and one of them turns out to be exactly the condition from part (i).

Solution

We model A, B, C as particles moving along one straight line on a smooth surface, with instantaneous collisions. All velocities are measured in the direction of A's initial motion.

Part (i)

The first collision. Momentum is conserved, and Newton's law of restitution says that the speed of separation is e times the speed of approach:

$$mu = mv_1 + kmv_2, \qquad v_2 - v_1 = eu.$$

Substituting $v_1 = v_2 - eu$ into the first equation gives $u = v_2 - eu + kv_2$, so

$$v_2 = \frac{1+e}{k+1}\,u = \beta u, \qquad v_1 = \frac{1+e}{k+1}\,u - eu = \frac{1+e-ek-e}{k+1}\,u = \frac{1-ke}{k+1}\,u = \alpha u,$$

as required.

The second collision. Measure time from the first collision. B covers the distance D to the wall in time $D/(\beta u)$. It rebounds with speed $e\beta u$, so it covers $\frac{1}{2}D$ back towards the collision point in a further time $D/(2e\beta u)$. If the second collision happens there, it happens at time

$$T = \frac{D}{\beta u}\left(1 + \frac{1}{2e}\right) = \frac{(2e+1)D}{2e\beta u}.$$

The point is $\frac{1}{2}D$ from the wall, so it is also $\frac{1}{2}D$ *beyond* the first collision point. A moves at the constant velocity αu, so it must have $\alpha > 0$ and $\alpha u T = \frac{1}{2}D$:

$$\alpha \frac{(2e+1)D}{2e\beta} = \frac{D}{2} \quad \Longleftrightarrow \quad \alpha(2e+1) = e\beta.$$

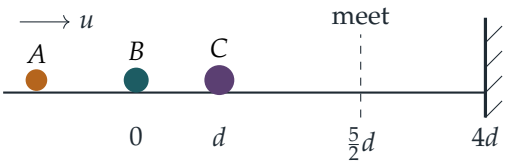
Multiplying by $k+1$ gives $(1-ke)(2e+1) = e(1+e)$, so

$$ke(2e+1) = (2e+1) - e - e^2 = 1 + e - e^2 \quad \Longrightarrow \quad k = \frac{1+e-e^2}{e(2e+1)},$$

which is the required result. This value does give $\alpha > 0$: it makes $ke = \frac{1+e-e^2}{2e+1}$, and this is less than 1 because $1 + e - e^2 < 1 + 2e$ for $e > 0$.

Part (ii)

Put the origin at B's starting point, so that C starts at $x = d$ and the wall is at $x = 4d$. The required meeting point is $\frac{3}{2}d$ from the wall, at $x = \frac{5}{2}d$.



Take $t = 0$ at the moment A hits B. That collision is exactly the one in part (i), so afterwards A moves with velocity αu and B with βu, where α and β are as before.

Ask yourself What happens when B hits C? Do we need to redo the collision algebra?

No. C is at rest and its mass $k^2 m$ is k times B's mass km. So this is the part (i) collision again, with mass ratio k, the same e, and incoming speed βu instead of u. B hits C at time $t_1 = d/(\beta u)$. Afterwards B moves at $\alpha\beta u$ and C at $\beta^2 u$.

Now we work out when each particle reaches $x = \frac{5}{2}d$, assuming no other collisions happen first (we check this at the end).

- A moves at αu from $x = 0$, so it gets there at $t = \dfrac{5d}{2\alpha u}$.
- B reaches $x = d$ at t_1 and then covers $\frac{3}{2}d$ at speed $\alpha\beta u$. It gets there at $t = \dfrac{d}{\beta u} + \dfrac{3d}{2\alpha\beta u}$.
- C covers $3d$ to the wall at $\beta^2 u$, then $\frac{3}{2}d$ back at $e\beta^2 u$. It gets there at

$$t = \frac{d}{\beta u} + \frac{3d}{\beta^2 u}\left(1 + \frac{1}{2e}\right) = \frac{d}{\beta u} + \frac{3(2e+1)d}{2e\beta^2 u}.$$

For a simultaneous collision all three times must be equal.

B **and** *C*. Cancel the common term $d/(\beta u)$ and multiply by $2\beta^2 u/(3d)$:

$$\frac{1}{\alpha}\beta = \frac{2e+1}{e} \quad \Longleftrightarrow \quad e\beta = \alpha(2e+1).$$

This is exactly the condition from part (i), so

$$k = \frac{1+e-e^2}{e(2e+1)}. \tag{1}$$

(This is no coincidence. In both situations, a particle leaves a collision, travels D to the wall at β times the incoming speed and comes back $\frac{1}{2}D$. Meanwhile the other particle travels $\frac{1}{2}D$ forward at α times the incoming speed. Here $D = 3d$, and the incoming speed is βu.)

A **and** *B*. Multiply $\dfrac{5}{2\alpha} = \dfrac{1}{\beta} + \dfrac{3}{2\alpha\beta}$ by $2\alpha\beta$ to get $5\beta = 2\alpha + 3$. Multiplying by $k+1$:

$$5(1+e) = 2(1-ke) + 3(k+1) \quad \Longrightarrow \quad 5e = k(3-2e) \quad \Longrightarrow \quad k = \frac{5e}{3-2e}. \tag{2}$$

Ask yourself We have two expressions for k. How do we get e?

We equate (1) and (2). Since $0 < e < 1$, both denominators are positive, so we can cross-multiply:

$$\begin{aligned} 5e^2(2e+1) &= (3-2e)(1+e-e^2) \\ 10e^3 + 5e^2 &= 3 + e - 5e^2 + 2e^3 \\ 0 &= 8e^3 + 10e^2 - e - 3. \end{aligned}$$

Trying $e = \frac{1}{2}$ gives $1 + \frac{5}{2} - \frac{1}{2} - 3 = 0$, so $2e - 1$ is a factor:

$$8e^3 + 10e^2 - e - 3 = (2e-1)(4e^2 + 7e + 3) = (2e-1)(4e+3)(e+1).$$

The other roots, $-\frac{3}{4}$ and -1, are negative, so the only root with $0 < e < 1$ is $e = \frac{1}{2}$, as required. Then $k = \frac{5}{4}$.

Checking the scenario is consistent. With $e = \frac{1}{2}$ and $k = \frac{5}{4}$ we get $\alpha = \frac{1}{6}$ and $\beta = \frac{2}{3}$. Before B hits C, A (speed $\frac{1}{6}u$) is slower than B (speed $\frac{2}{3}u$), so it cannot catch B early. After that, B (speed $\frac{1}{9}u$) is slower than C (speed $\frac{4}{9}u$), so B cannot catch C before C reaches the wall. B is now slower than A, so A gains on B, and they meet only at the point we computed. So there are no extra collisions, and all three particles reach $x = \frac{5}{2}d$ at time $t = 15d/u$.

Checks

- In part (i), $v_1 + kv_2 = u$ (momentum) and $v_2 - v_1 = eu$ (restitution). If $e = 1$ and $k = 1$, then $\alpha = 0$ and $\beta = 1$: equal masses swap velocities in a perfectly elastic collision, as they should.

- In part (ii), with $u = d = 1$ and the values above, the three arrival times are $\frac{5/2}{1/6} = 15$ for A, $\frac{1}{2/3} + \frac{3/2}{1/9} = \frac{3}{2} + \frac{27}{2} = 15$ for B, and $\frac{3}{2} + \frac{3}{4/9} + \frac{3/2}{2/9} = \frac{3}{2} + \frac{27}{4} + \frac{27}{4} = 15$ for C.
- Formula (1) at $e = \frac{1}{2}$ gives $k = \frac{5/4}{1} = \frac{5}{4}$, which agrees with (2): $\frac{5/2}{2} = \frac{5}{4}$.

Common slips

- Measuring the meeting point in part (i) from the wall when you need it from the collision point. Here they happen to be the same, $\frac{1}{2}D$, but in part (ii) the meeting point $\frac{3}{2}d$ from the wall is $\frac{5}{2}d$ from B's start.
- Redoing the B–C collision from scratch and losing a factor. Noticing that the mass ratio is again k saves the work.
- Forgetting that C comes back from the wall at $e\beta^2 u$ rather than $\beta^2 u$.

Beyond STEP

This question works because the problem is *self-similar*: scaling the masses by k and the speed by β turns one collision into the next, so the same algebra repeats. The same idea is behind the chains of balls of geometrically decreasing mass that pass energy along very efficiently, and behind the famous "counting π with colliding blocks" problem, where the analysis is cleanest in velocity space with momentum and energy as constraints. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions systematically, including working in the centre-of-mass frame, which makes formulas like α and β easy to see.

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